

Tables for editor

Observed uptake, geographic reach, and patient-flow structure of therapeutic ERCP in Brazil's Unified Health System, 2021-2025

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Table 1. Study cohorts and analytical population.

Measure	Value	Denominator	Evidence	Key limitation
Cohort A unique AIHs	91,175 AIHs	—	descriptive	Billing AIHs are deduplicated administrative records, not independently adjudicated clinical cases.
Cohort B final unique AIHs	33,497 AIHs	91,175 Cohort A unique AIHs	descriptive	Administrative diagnosis fields do not provide chart-adjudicated phenotype validation.
Hospitals with observed coded therapeutic ERCP use	445 hospitals	—	descriptive	This is observed uptake, not a true technology-adoption date.
Aim 4 complete primary analysis rows	30,900 AIHs	33,497 Cohort B unique AIHs	associational	Complete-context restriction and residual confounding limit generalisability and causal interpretation.
Aim 4 hospitals	360 hospitals	30,900 Aim 4 analysis AIHs	associational	Hospital-level associations remain susceptible to residual confounding and referral selection.

AIH, Autorização de Internação Hospitalar; ERCP, endoscopic retrograde cholangiopancreatography.

Table 2. Observed therapeutic ERCP use and maintenance by hospital.

Measure	Value	Denominator	Evidence	Key limitation
Hospitals with observed coded therapeutic ERCP use	445 hospitals	—	descriptive	This is observed uptake, not a true technology-adoption date.
Prevalent users at the January 2021 boundary	23 hospitals	445 hospitals ever observed	descriptive	The observation boundary cannot distinguish earlier use from first use.
First observed coded use in 2021	176 hospitals	445 hospitals ever observed	descriptive	First observed coded use is not a true adoption date.
First observed coded use in 2022	83 hospitals	445 hospitals ever observed	descriptive	First observed coded use is not a true adoption date.
First observed coded use in 2023	66 hospitals	445 hospitals ever observed	descriptive	First observed coded use is not a true adoption date.
First observed coded use in 2024	63 hospitals	445 hospitals ever observed	descriptive	First observed coded use is not a true adoption date.
First observed coded use in 2025	57 hospitals	445 hospitals ever observed	descriptive	First observed coded use is not a true adoption date.
Maintained service through first completed 12 months	252 hospitals	390 first-12-month evaluable hospitals	descriptive	Maintenance is defined from coded billing continuity rather than audited service capacity.
First-observed-use model status	PASS WITH WARNING	—	associational	Events per parameter were below the conventional 10-per-parameter heuristic; results are associational.

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Supplementary Table 1. Data sources and acquisition boundaries.

Source	Custodian	Analytical role	Sharing boundary
SIH/SUS hospitalisation and procedure-detail records	DATASUS, Ministry of Health	AIH unit, procedure code, diagnoses, residence, provider, in-hospital disposition	Restricted individual rows are not redistributed
CNES-PF	DATASUS	Time-aligned hospital eligibility and capacity context	Public-source acquisition instructions only
Population and municipality geometry	IBGE	Adult denominators, municipal identifiers, geometry, audited city-seat anchors	Publicly documented acquisition route
IVS 2010	Ipea-linked official source	Static municipal ecological context	No individual socioeconomic interpretation
Supplementary-insurance coverage	ANS	Municipal contextual measure	Contextual, not individual insurance status
OpenStreetMap extract	OpenStreetMap contributors	Road graph (brazil-260822.osm.pbf)	Licence and acquisition route documented
GraphHopper	Local reproducible service	ercp_car route and contour profile	Version/configuration evidence must be mirrored before final integrity
CONITEC, Portarias, SIGTAP	Ministry of Health	Policy and code chronology	Official sources and access dates documented

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Supplementary Table 2. Cross-aim estimands, denominators, and analysis boundaries.

Component	Cohort / unit	Denominator	Estimand or descriptive target	Inference level	Missingness / unknown rule	Reporting status
Aim 1 observed uptake	Cohort hospitals A	445 hospitals ever observed	First month of observed coded therapeutic ERCP use	Descriptive	January 2021 active hospitals are left-censored prevalent users	Reported
Aim 1 maintenance	Cohort hospitals A	390 with a completed first 12-month window	Activity in at least 6 months of the first completed 12-month window	Descriptive	Incomplete windows excluded from the risk set	252/390 (64.6%); reported
Aim 1 first-observed-use model	Eligible hospital-months	Frozen CNES risk set	Associational model of first observed coded use	Associational	No causal adoption interpretation	Associational; limited events per parameter
Aim 2 ecological equity	Cohort municipality-years B	27,825 municipality-years	Adult-standardised treated-use gradient across ranked IVS	Ecological associational	Static 2010 IVS; no individual interpretation	SII and RII reported; modified-Poisson inference DOWNGRADE
Aim 2 realised routes	Cohort B routed AIHs	33,321 routed AIHs	Flow-weighted route-time distribution among treated flows	Descriptive	Unrouted flows excluded from route summaries; no unmet-need inference	Reported
Aim 2 potential reach	Adult municipality-year population	801,100,903 adult person-years in anchor audit	Population within 120 and cumulative 180 road-minutes of observed provider municipalities	Descriptive opportunity modelled	Invalid anchors remain unknown; one municipality unknown in 2025	National component reported; regional and vulnerability-gap components NOT_EVALUATED
Aim 3 patient-flow network	Observed residence-to-provider flows and pooled municipal projection	3,338 nodes; 166,959 shared-destination edges	Destination and shared-destination structure	Descriptive	Public cells below 5 suppressed; no patient-to-patient interpretation	Reported
Aim 3 service areas	Residence municipality-year records	5,565 municipal IVS records for contextual summaries	Modal observed destination and destination share	Descriptive	Suppressed flows remain suppressed; no need or appropriateness inference	Reported
Aim 3 structural resilience	Potential-access matrix / adults	8,723,036 matrix rows; 142,299 reachable rows	Adults losing an alternative provider under targeted hub removal	Descriptive structural stress test	Complete-case and upper-bound handling retain unknown anchors	Reported as a structural stress test; not a real-world counterfactual
Aim 4 supportive outcome	Cohort complete-context AIHs B	30,900 AIHs; 378 deaths; 360 hospitals	Standardised risk contrast at hospital-volume p90 versus p10	Associational/supportive	Complete-context design; no imputation-based alternative was used	Bootstrap DOWNGRADE; formal CI NOT_EVALUATED
Planned staggered-uptake analysis	Hospital-month design	No pre-first-observed-use hospital-months	Planned quasi-causal family	Not a release result	Structural absence of pre-period observations	Not reported because pre-period observations were unavailable

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Supplementary Table 3. Ecological equity, travel, potential reach, and network measures.

Measure	Value	Denominator	Evidence	Key limitation
IVS relative index of inequality	0.55	27,825 municipality-years	associational	IVS is an ecological contextual exposure and does not identify individual socioeconomic effects.
IVS slope index of inequality	-2.56 treated AIHs per 100,000 adults	27,825 municipality-years	associational	IVS is an ecological contextual exposure and the association is not causal.
Flow-weighted median realised route time	23.2 minutes	33,321 AIHs with a route	descriptive	Municipal anchors approximate residence and provider locations; this is realised treated flow, not direct patient travel.
Flow-weighted 90th percentile realised route time	213.4 minutes	33,321 AIHs with a route	descriptive	Municipal anchors approximate residence and provider locations; this is realised treated flow, not direct patient travel.
Realised treated flow above 120 minutes	20.7%	33,321 AIHs with a route	descriptive	This describes routed treated flows and cannot be interpreted as population access or unmet need.
Adult population within 120 road-minutes in 2021	71.9%	157,421,555 adults	descriptive	Modelled potential road access does not measure realised access, capacity, quality, or need.
Adult population within 180 road-minutes in 2021	81.2%	157,421,555 adults	descriptive	The cumulative 180-minute estimand unions raw 120- and 180-minute contours when GraphHopper approximation is non-nested.
Adult population within 120 road-minutes in 2025	82.1%	163,078,720 anchored adults	descriptive	One 2025 municipality lacks a road anchor; anchored-denominator and bounds must be reported.
Adult population within 180 road-minutes in 2025	90.4%	163,078,720 anchored adults	descriptive	One 2025 municipality lacks a road anchor; the cumulative geometry rule and bounds must be retained.
Adult population road-anchor coverage	99.9995%	801,100,903 adult person-years	descriptive	Municipalities without anchors remain unknown and are never recoded as uncovered.
Aim 2 primary-family inferential status	DOWNGRADE	—	associational	Modified-Poisson standardized potential-coverage predictions exceeded probability 1; the model was not rerun or relaxed.
Pooled patient-flow projection nodes	3,338 municipalities	—	descriptive	The patient-flow network reflects observed treated flows, not population need or referral appropriateness.
Pooled patient-flow projection edges	166,959 edges	—	descriptive	Projection edges encode shared performing destinations and are not direct patient-to-patient links.
Potential-access matrix rows	8,723,036 rows	—	descriptive	This is a modelled potential-access matrix and contains no capacity or realised-use inference.
Reachable potential-access matrix rows	142,299 rows	8,723,036 potential-access matrix rows	descriptive	Reachability is potential road access, not observed service use or capacity.
IVS municipal-total records selected	5,565 municipalities	—	descriptive	The 2010 IVS is a static ecological contextual exposure carried across 2021-2025.

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Supplementary Table 4. Supportive outcome-analysis measures and inferential limitations.

Measure	Value	Denominator	Evidence	Key limitation
Aim 4 complete primary analysis rows	30,900 AIHs	33,497 Cohort B unique AIHs	associational	Complete-context restriction and residual confounding limit generalisability and causal interpretation.
Aim 4 in-hospital death events	378 deaths	30,900 Aim 4 analysis AIHs	associational	Administrative in-hospital death does not capture post-discharge mortality.
Aim 4 hospitals	360 hospitals	30,900 Aim 4 analysis AIHs	associational	Hospital-level associations remain susceptible to residual confounding and referral selection.
Aim 4 bootstrap validity status	DOWNGRADE	2,000 expected replicates	associational	The prespecified bootstrap validity criterion was not met; no reduced model or substitute interval was used.
Valid AS_mean bootstrap replicates	618 replicates	2,000 expected replicates	associational	Valid-replicate yield was below the prespecified threshold; inferential intervals are downgraded.
Valid MPL_Jeffreys bootstrap replicates	617 replicates	2,000 expected replicates	associational	Valid-replicate yield was below the prespecified threshold; inferential intervals are downgraded.
Planned quasi-causal analysis	Not reported	—	associational	No pre-first-observed-use hospital-months were available; quasi-causal estimates and negative controls are not reported.

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Supplementary Table 5. RECORD reporting checklist.

RECORD item group	Locator	Status
Routinely collected data source and setting	Methods: Data sources; the supplementary data-sources section	Reported
Code/algorithm-based population selection	Methods: Cohorts; the supplementary cohort-construction section	Reported
Validation of codes/algorithms	Discussion limitations; the supplementary cohort-construction section	No chart validation available; explicitly stated
Linkage methods and quality	Methods: Cohorts; the supplementary cohort-construction section	Exact linkage rule reported; reason-specific intermediate breakdown unavailable
Data cleaning and deduplication	Methods: Cohorts; the supplementary cohort-construction section	Reported
Data access and privacy	Data availability; the supplementary sections 'Data sources and acquisition boundary' and 'Aim 3 patient-flow network, service areas, privacy, and resilience'	Reported
Flow of records and exclusions	Figure 1; the supplementary sections 'Code lists, linkage, and cohort construction' and 'Variable availability and missing-data handling'	Aggregate flow reported; reason-specific intermediate breakdown unavailable
Extent of database access for replication	Code/Data availability; the supplementary reproducibility and declarations section	Reported with restricted-row boundary

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Supplementary Table 6. STROBE reporting checklist.

STROBE item	Locator	Status
Title/abstract design	Title; Structured abstract	Reported
Background and objectives	Introduction	Reported
Study design, setting, dates	Methods: Study design	Reported
Participants and eligibility	Methods: Cohorts; the supplementary cohort-construction section	Reported
Variables, outcomes, exposures, confounders	Methods by aim; the supplementary estimand map and Aim 4 section	Reported
Data sources and measurement	Methods: Data sources; the supplementary sections on data sources and the aim-specific methods	Reported
Bias	Methods: Missing data; Discussion limitations	Reported
Study size	National administrative census; the supplementary estimand map	Reported; no sampling power calculation used
Quantitative variables and statistical methods	Methods by aim; the supplementary estimand map and aim-specific sections	Reported
Missing data	Methods; the supplementary missing-data section	Reported within frozen local evidence limits
Participants and descriptive data	Results; Figures 1–2; Tables 1–2	Reported
Outcome and main results	Results; Figures 3–6, Supplementary Figure 1, and Supplementary Tables 5–6	Reported with downgrades
Other analyses	the supplementary aim-specific and quasi-causal eligibility sections	Reported; planned quasi-causal estimates excluded
Key results, limitations, interpretation, generalisability	Discussion	Reported
Funding	Declarations	Pending author metadata

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